

Meteoroids and Space Debris

11.1 Introduction

Meteoroids and space debris move at sufficient relative speeds that hypervelocity impacts constitute a serious hazard to spaceflight as they can damage or destroy spacecraft. This is of particular concern for crewed spacecraft. As described in Chapter 2, a *meteoroid* is a small piece of celestial material orbiting the sun from a comet or asteroid, where from the latter it is often from the collisions of asteroids. A *micrometeoroid* is a small meteoroid with a mass typically less than a gram. *Orbital debris* is manmade objects in orbit that are not carrying out a useful function. Examples of orbital debris include spent rockets, clamps, despin cables, metallic fragments, paint, rocket exhausts, and remnants of launch vehicle and spacecraft explosions.

Many of the spacecraft used for navigation, communication, and weather assessment share similar orbital altitudes. Popular orbits are low Earth orbits (LEO) on the order of 1000-km altitude, semisynchronous orbits at $\approx 20,200$ -km altitude, and geosynchronous equatorial orbits (GEO) at $\approx 35,800$ -km altitude. These specific orbital trajectories allow the spacecraft to carry out their missions effectively. Because of the popularity of these orbits, there is an increase in space debris at these altitudes. Spacecraft remain in orbit until atmospheric drag causes them to reenter the Earth's atmosphere. Orbital lifetimes are difficult to predict because of the inability to predict accurately solar and geomagnetic activities, errors inherent in models of atmospheric density, and uncertainty in area to mass ratios and drag coefficients. Debris above 1000 km can remain in orbit for several thousands of years, and since atmospheric density decreases with increasing altitude, lifetimes increase with increasing altitudes, as illustrated in Fig. 11.1. Since the area-to-mass ratio is typically proportional to l^{-1} , where l is the characteristic dimension of an object, larger objects have smaller area-to-mass ratios and will remain in orbit longer. Conversely, smaller objects typically reenter the Earth's atmosphere sooner than larger objects.

Orbital bodies that are cataloged and tracked periodically so that their orbits are determined can be subdivided into five categories: payloads, rocket bodies, mission-related debris, breakup debris, and anomalous debris, as illustrated in Table 11.1.

Payloads consist of functional and nonfunctional spacecraft. *Rocket bodies* are the rockets used to launch payloads. *Mission-related debris* is the intentional

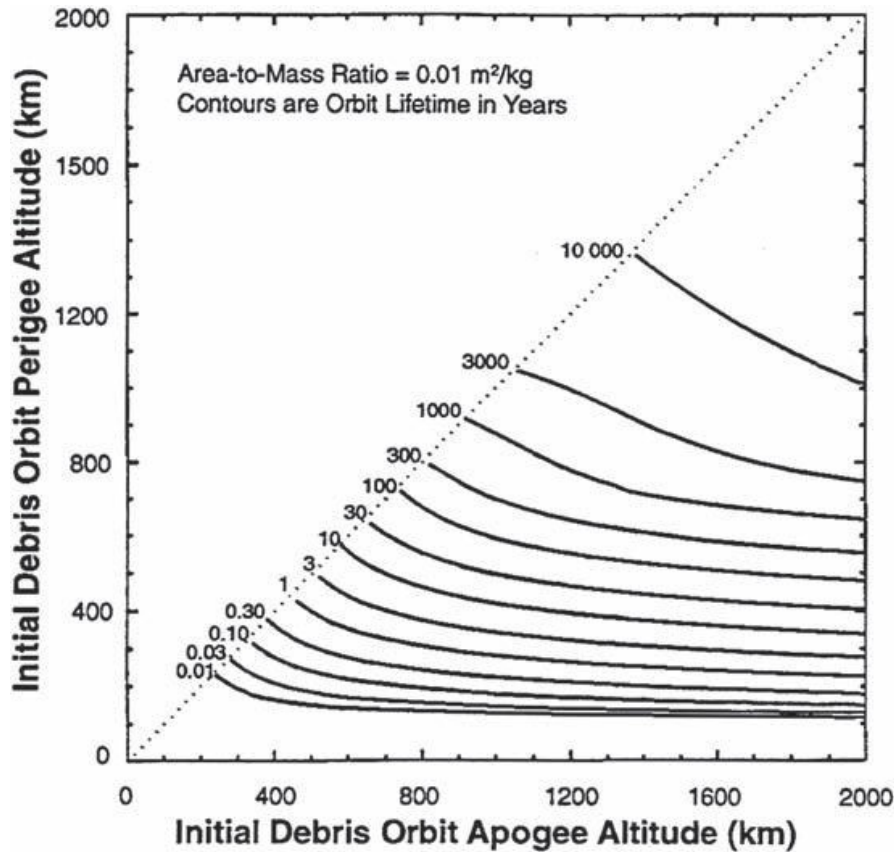


Fig. 11.1 Debris orbital lifetimes. [From NASA NSS 1740.14 (NASA 1995); courtesy of NASA.]

separation of objects from payloads. *Breakup debris* is intentional or unintentional fragmentation of an orbiting body, often by explosion of the propulsion system. *Anomalous debris* is the unplanned separation of an object from a payload, often caused by material degradation such as thermal blankets or solar panels.

Impacts of meteoroids and debris with the shuttle have created damage of windows, radiators, and other components, as illustrated in Table 11.2 of the 20 most significant shuttle impacts.

Several collisions of objects in orbit have occurred. In December 1991, a non-functional Russian navigation satellite, Cosmos 1934 (1988-023A, U.S. Satellite

Table 11.1 Categorization of cataloged orbital population

Cause	Percent
Payloads	31.3
Rocket bodies	16.6
Mission-related debris	13.0
Breakup debris	37.7
Anomalous debris	1.3

Source: From Whitlock (2004).

Table 11.2 Most severe 20 meteoroid and debris impacts on the shuttle from STS-50 through STS-86

<i>Windows</i>						
Mission no.	Duration, days	Window location	Flaw diameter, mm	Crater depth, mm	Particle type, SEMA/EDXA results	Estimated particle diameter, mm
STS-50	13.8	#4, RH forward	7.2	0.57	Orbital debris: Ti metal	0.20
STS-59	11.2	#11, side hatch	12.0	0.57	Orbital debris: spacecraft paint	0.22
STS-94	15.7	#7, RH overhead	8.2	0.55	Orbital debris: metallic Al	0.24
<i>Radiators</i>						
Mission no.	Duration, days	Radiator location	Tape hole diameter, mm	Facesheet hole diameter, mm	Particle type SEMA/EDXA results	Estimated particle diameter, mm
STS-50	13.8	LH#1	3.8	1.1	Orbital debris: spacecraft paint	0.5
STS-73	15.9	LH#4	8.3	1.1	Orbital debris: spacecraft paint	0.6
STS-79	10.1	RH#3	4.8	1.0	Orbital debris: stainless steel	0.1
STS-80	17.7	RH#4	5.5	2.8	Orbital debris: stainless steel	1.7
STS-80	17.7	LH#4	3.2	2.0	Orbital debris: stainless steel	1.0
STS-81	10.1	RH#4	4.3	1.5	Orbital debris: stainless steel	0.3
STS-84	9.2	RH#4	4.0	Unknown	Orbital debris: stainless steel	0.2
STS-85	11.9	RH#4	5.0	1.3	Meteoroid	0.7
STS-84	10.8	Ext. manifold-1	9.0	0.5 depth	Orbital debris: stainless steel	0.4
STS-86	10.8	Ext. manifold-2	1.0	0.4 depth	Meteoroid	0.2

(Continued)

Table 11.2 Most severe 20 meteoroid and debris impacts on the shuttle from STS-50 through STS-86 (continued)

<i>Other orbiter components and payloads</i>						
Mission no.	Duration, days	Impact location	Damaged material	Hole diameter, mm	Particle type, SEMA/EDXA results	Estimated particle diameter, mm
STS-55	10.0	DEA box of Ku-band antenna	Ag-Teflon tape on Al	4.1	Orbital debris; metallic Al	0.3
STS-56	9.3	Reflector of Ku-band antenna	Graphite-epoxy	1.4	Meteoroid	0.6
STS-72	8.9	Thermal spring seal of rudder speed brake	Inconel, RTV	3.4	Orbital debris; metallic Al	1.3
STS-73	15.9	FRSI ^a ext. PLB door LH #4	Nomex felt	17	Orbital debris; circuit board components	3 L × 1 diam
STS-75	15.7	Tethered satellite system pallet trunnion	Titanium	1.0	Orbital debris; metallic Al	0.8
STS-84	9.2	FRSI ext. PLB door RH #2	Nomex felt	12	Orbital debris; metallic Al	2.1
STS-94	15.7	Conical seal of vertical stabilizer	Inconel	0.9	Meteoroid	0.4

^aFRSI = flexible reusable surface insulation.
Source: From Christiansen (2003); courtesy of NASA.

Number 18985), collided with a piece of debris from a sister spacecraft, Cosmos 926. Both objects were in similar orbits with a mean altitude of 980 km and an inclination of 83 deg. Two pieces of debris from Cosmos 1934 (1988-023C, U.S. Satellite Number 21912, and 1988-023D, U.S. Satellite Number 22919) were discovered to have collided by the Space Surveillance Network within a few weeks of the event. On 24 July 1996 there was a collision between the French spacecraft Cerise with debris from the third stage of an Ariane rocket. The spacecraft lost a section of its gravity gradient boom that left it seriously damaged. On 17 January 2005, a 31-year-old U.S. spent rocket (1974-015B, U.S. Satellite Number 07219) and a fragment (1999-057CV, U.S. Satellite Number 26207) from the third stage of a Chinese CZ-4 launch vehicle, which exploded in March 2000, collided. The collision occurred at an altitude of 885 km above the South Polar Region. Both objects were in similar retrograde orbits at the time of the collision. The orbits of both objects were disturbed and three pieces of debris (cataloged as U.S. Satellite Numbers 28591–28593) were released from the U.S. rocket.

11.2 Meteoroids and Debris Observations

Characteristics of meteoroids and space debris are determined by several means, including meteorites (remnants of meteoroids that reach the Earth's surface), radars, telescopes, retrieved spacecraft, and lunar meteorites returned by the Apollo program. Radars in use operate in various modes: stare mode, track mode, or mixed mode. In the stare mode, the radar's orientation is fixed. In the track mode, the radar tracks an object to increase the accuracy of its observations. In the mixed mode, the radar is initially in a stare mode but follows an object that it detects if it deems it necessary to increase accuracy. As examples, the bistatic radar at Effelsberg has a sensitivity of 9 mm at 1000-km, range, the Goldstone bistatic radar has an accuracy of 2 mm at 1000 km, and the Haystack monostatic radar has a sensitivity of 6 mm at 1000 km (United Nations 1999). Optical instruments have the ability to detect objects as small as 60 cm at geostationary altitudes. Retrieved spacecraft and spacecraft components include the Long Duration Exposure Facility (LDEF) spacecraft (5.8 years in orbit), the solar arrays from the Hubble Space Telescope, the Solar Maximum spacecraft (4.1 years in orbit), and inspections of the shuttle after each flight. For example, the LDEF had over 30,000 craters visible to the naked eye, of which 5000 had diameters larger than 0.5 mm.

Figure 11.2 gives an estimate of the meteoroid and debris flux as a function of the particle diameter obtained from the variety of sources identified.

11.3 Meteoroid Environment and Models

11.3.1 Introduction

Meteoroid models that can be used in the design of both spacecraft and human exploration missions are available in both analytical and digital computer format. Analytic models include the Grün et al. (1985) analytic model, the Cour-Palais (1969) meteoroid stream model, and the Jenniskens (1994) meteoroid stream model. Digital computer meteoroid models include the ESA Meteoroid Model (Dakarev et al. 2005) and the NASA Meteoroid Engineering Model (McNamara et al. 2004).

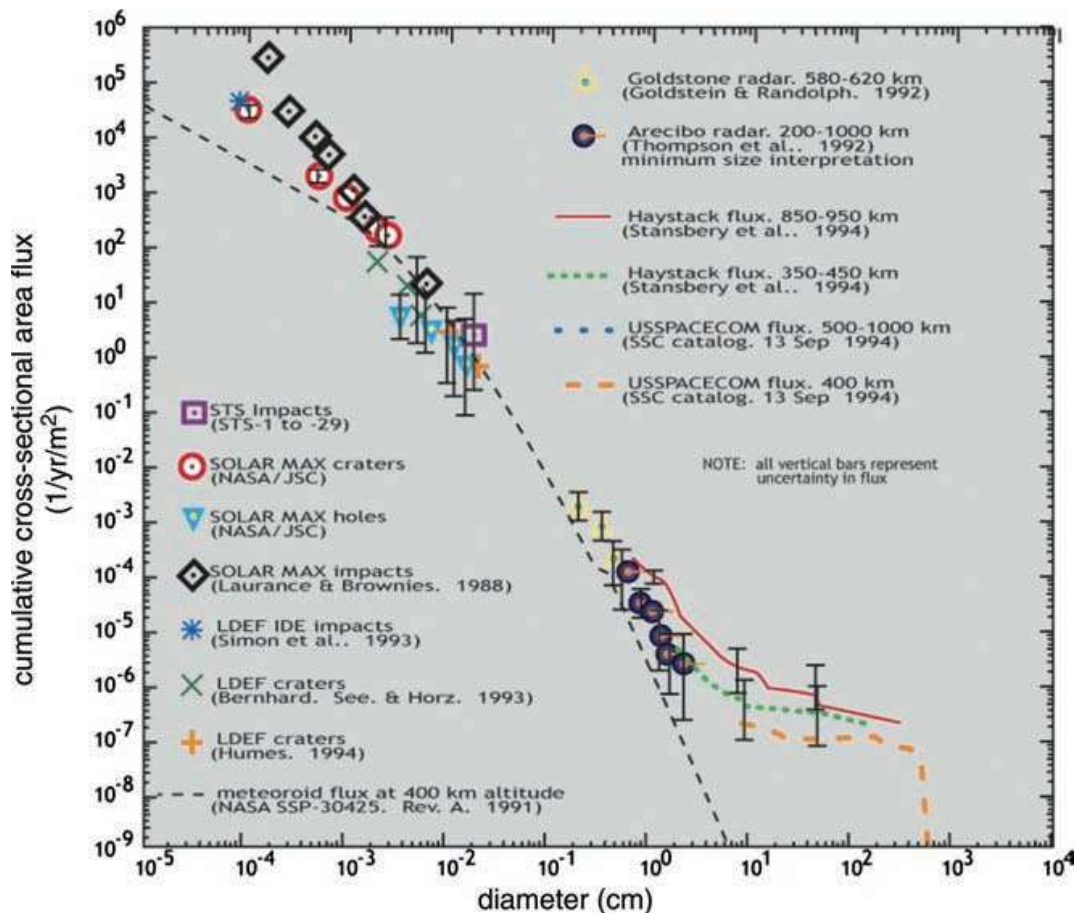


Fig. 11.2 Meteoroids and space debris flux as a function of diameter. [From Heimerdinger (2005); copyrighted 2005, IEEE.]

11.3.2 Grün Meteoroid Model

The Grün et al. (1985) interplanetary model for meteoroids gives the integral meteoroid flux averaged over the year in terms of the number of particles per unit area per unit time for mass greater than a specified value that impact a flat plate over a solid angle of 2π steradians at a distance of 1 AU from the sun. The average flux $F_c(m)$ in the vicinity of the Earth as described in SPENVIS (Space Environment Information System 2007) and Anderson and Smith (1994) is given by

$$F_c(m, h) = F(m) \chi_i(h) G_e(h) \quad (11.1)$$

where

$F_c(m, h)$ = average flux at 1 AU from the sun in vicinity of the Earth with mass $\geq m$, particles $\text{m}^{-2} \text{y}^{-1} (2\pi \text{ sr})^{-1}$

$F(m)$ = average flux at 1 AU from the sun with mass $\geq m$, particles $\text{m}^{-2} \text{y}^{-1} (2\pi \text{ sr})^{-1}$

$\chi_i(h)$ = Earth shielding factor, dimensionless

$G_e(h)$ = gravitational focusing factor, dimensionless

h = spacecraft altitude, m

11.3.2.1 Flux at 1 AU. The flux at 1 AU from the sun is given by

$$\begin{aligned} F(m) &= 3.15576 \times 10^7 [F_1(m) + F_2(m) + F_3(m)] \\ F_1(m) &= (2.2 \times 10^3 m^{0.306} + 15.0)^{-4.38} \\ F_2(m) &= 1.3 \times 10^{-9} (m + 10^{11} m^2 + 10^{27} m^4)^{-0.36} \\ F_3(m) &= 1.3 \times 10^{-16} (m + 10^6 m^2)^{-0.85} \end{aligned} \quad (11.2)$$

where

$F_1(m)$ = refers to particles with $10^{-9} \text{ g} < m < 1 \text{ g}$
 $F_2(m)$ = refers to particles with $10^{-14} \text{ g} < m < 10^{-9} \text{ g}$
 $F_3(m)$ = refers to particles with $10^{-18} \text{ g} < m < 10^{-14} \text{ g}$
 m = mass, g

11.3.2.2 Earth shielding. If the surface under study is near the Earth, as on an Earth satellite, the meteoroid flux can be partially shielded by the Earth, reducing the average flux density. On the assumption that the flux is omnidirectional, three conditions are proposed for the Earth shielding factor:

$$\chi_1 = 1 \quad \text{if the angle between the normal to the surface and the direction of the Earth is } \geq \pi/2 \quad (11.3)$$

$$\chi_2 = \frac{1}{2} (1 + \cos \theta) \quad \text{if the direction of the normal to the surface has a random orientation} \quad (11.4)$$

$$\chi_3 = \cos \theta \quad \text{if the normal to the surface is pointing toward the center of the Earth} \quad (11.5)$$

where

h_a = height of atmosphere, $\approx 100,000 \text{ m}$

R_e = radius of the Earth, km

$$\sin \theta \equiv \frac{R_e + h_a}{R_e + h} \quad (11.6)$$

and the geometry is illustrated in Fig. 11.3.

The geometrical shielding factor is illustrated in Fig. 11.4 for the three cases identified in Eqs. (11.3–11.5).

11.3.2.3 Gravitational focusing. The gravitational field of the Earth attracts or focuses meteoroid particles so that the flux in the vicinity of the Earth is increased by this phenomenon by the gravitational focusing factor

$$G_e(h) = 1 + \frac{R_e + h_a}{R_e + h} \quad (11.7)$$

which is illustrated in Fig. 11.5.

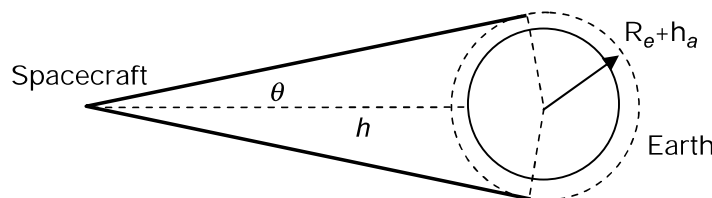


Fig. 11.3 Geometry for Earth shielding.

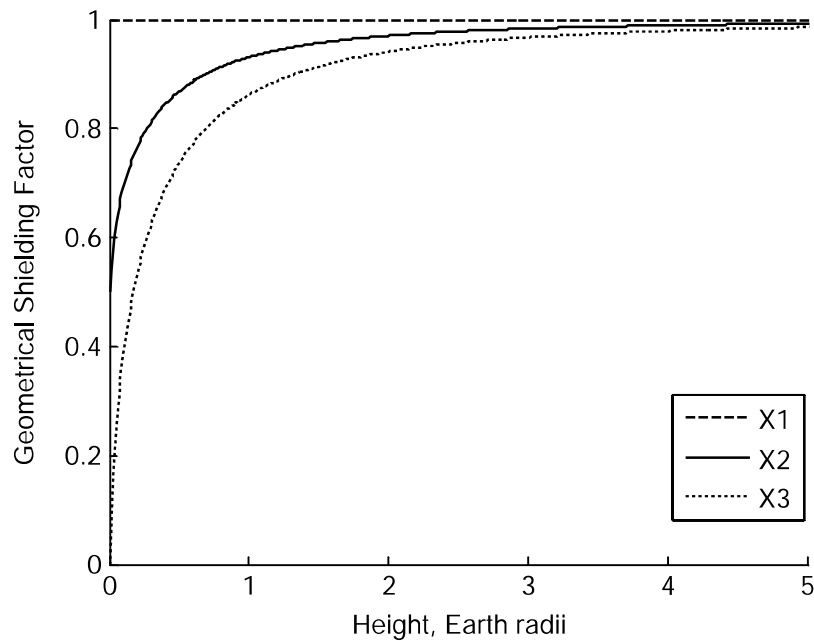


Fig. 11.4 Geometrical shielding factors, Eqs. (11.3–11.5).

11.3.3 Meteoroid Speed Distribution

Several estimates are available for the speed of meteoroids. The speed probability density distribution of meteoroids with respect to the Earth is given in NASA TM 4527 by Anderson and Smith (1994) in terms of $n(v)$, the fraction of particles with specified speed, as

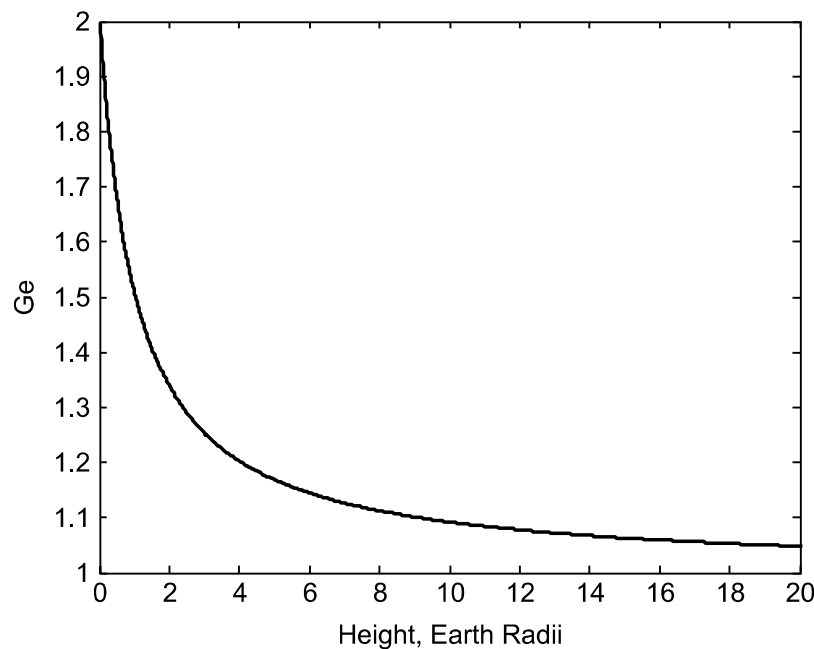


Fig. 11.5 Gravitational focusing factor, Eq. (11.7).

$$\begin{aligned}
 n(v) &= 0.112 & 11.1 \leq v < 16.3 \text{ km s}^{-1} \\
 n(v) &= 3.328 \times 10^5 v^{-5.34} & 16.3 \leq v < 55 \text{ km s}^{-1} \\
 n(v) &= 1.695 \times 10^{-4} & 55 \leq v < 72.2 \text{ km s}^{-1}
 \end{aligned} \tag{11.8}$$

This distribution has an average omnidirectional speed of 15.6 km s^{-1} with respect to the Earth. The relative speed with respect to an orbiting spacecraft is dependent on the relative orientation of the spacecraft's surface with respect to the orbital velocity vector. On the ram side the relative speed would be the average meteoroid speed plus the orbital speed, while the wake side would have a relative speed of the average meteoroid speed minus the orbital speed.

11.3.4 Cour-Palais Meteoroid Model

Another meteoroid speed estimate, given by Cour-Palais (1969) and NASA SP-8013 (NASA 1969), is in histogram form with a mean value of 20 km s^{-1} and a range of $9\text{--}75 \text{ km s}^{-1}$. The probability density function as a function of relative speed given in NASA TM 4527 (Anderson and Smith 1994) and NASA SP-8013 (NASA 1969) is illustrated in Fig. 11.6.

11.3.5 Meteoroid Mass Uncertainty

There is uncertainty in the properties of the meteoroid environment near Earth, primarily from uncertainties in the particle densities and masses. According to Anderson and Smith (1994), meteoroid mass densities have values between 0.15 and 8.0 g cm^{-3} , which implies an uncertainty in flux of a factor of $0.1\text{--}10$. For meteoroids with mass less than 10^{-6} g , the mass is uncertain to within a factor of $0.2\text{--}5$ times the estimated value, which implies a flux

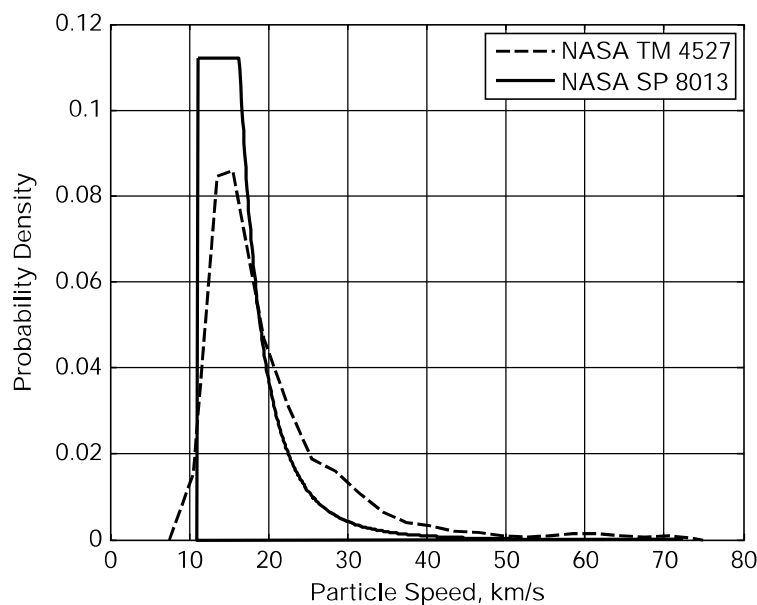


Fig. 11.6 Probability of meteoroids as a function of velocity.

uncertainty of a factor of 0.33–3 for a given mass. Uncertainties at other distances from the sun are larger.

11.3.6 Meteoroid Streams

Meteoroid streams are remnants of periodic comets or in rarer cases asteroids as described in Chapter 2. Models of the annual streams are available that give their times of occurrence, duration, maximum flux ratio, rates of increase and decrease in the flux rates, and velocity. Some 18 streams are characterized in NASA SP-8013 (NASA 1969) and 49 are characterized by Jenniskens (1994).

11.3.7 ESA Meteoroid Model

The ESA Meteoroid Model is a statistical digital computer model of meteoroids originating from comets and asteroids. It is based on the physics of their release and distribution and utilizes data from ground-based and spacecraft observations (Dikarev et al. 2005). It can be used to predict the meteoroid flux and speed of particles between a micron and several centimeters in size in the region of distances of 0.1 to 10 AU from the sun. The model incorporates the gravity of the planets, the Poynting-Robertson effect (a process in which solar radiation causes dust grains in the solar system to slowly spiral inward toward the sun); data from the COBE, Galileo, and Ulysses spacecraft instruments; and rock samples returned from the moon by the Apollo program.

11.3.8 NASA Meteoroid Engineering Model (MEM)

The NASA Meteoroid Engineering Model (MEM) is a digital computer model that provides estimates of the meteoroid flux in interplanetary space between 0.2 and 2.0 AU from the sun (McNamara et al. 2004). This model provides fluxes, speeds, and directionality of the meteoroids. Given a state vector, the model provides the flux and impact speeds on the surface of a cubic structure with one face in the ram direction.

11.4 Debris Environment and Models

11.4.1 Introduction

It is important that models of the debris environment are available so that spacecraft can be designed appropriately to withstand or avoid the potential impact of space debris. Several models are available, including the NASA Orbit Debris Engineering Model (ORDEM) and the ESA Meteoroid and Space Debris Terrestrial Environmental Reference Model (MASTER).

11.4.2 NASA Orbit Debris Engineering Model (ORDEM)

ORDEM is a computer model that provides information on debris of sizes from 10 μm to 10 m between the altitudes of 200–2000 km (Liou et al. 2002).

ORDEM is a series of models with the latest identified as ORDEM 2000. The model provides information for both designers of spacecraft and observers of debris. For spacecraft, the input is the orbital characteristics (semimajor axis, eccentricity, inclination, argument of perigee and right ascension of the ascending node), duration of mission, and the number of orbital segments. The outputs are tables and plots of the average speed and the integral average mass flux as a function of particle diameter, as illustrated in Fig. 11.7.

Figure 11.8 illustrates a comparison between ORDEM96, ORDEM2000, and the Haystack radar observations.

11.4.3 ESA Meteoroid and Space Debris Terrestrial Environmental Reference Model (MASTER)

The ESA MASTER meteoroid and debris model is a digital computer model that covers altitudes up to and above geostationary altitudes and has a lower size threshold of $1\ \mu\text{m}$. Several versions have been developed with the latest update MASTER-2005 (Oswald et al. 2005). Flux can be determined relative to an orbiting spacecraft or a fixed volume with respect to the Earth. Flux contributions can be segregated according to the debris source, such as cataloged objects, explosion and collision fragments, rocket motor slag, etc. Information provided includes the mass, dimensions, speed, and direction. The mean number of debris particles

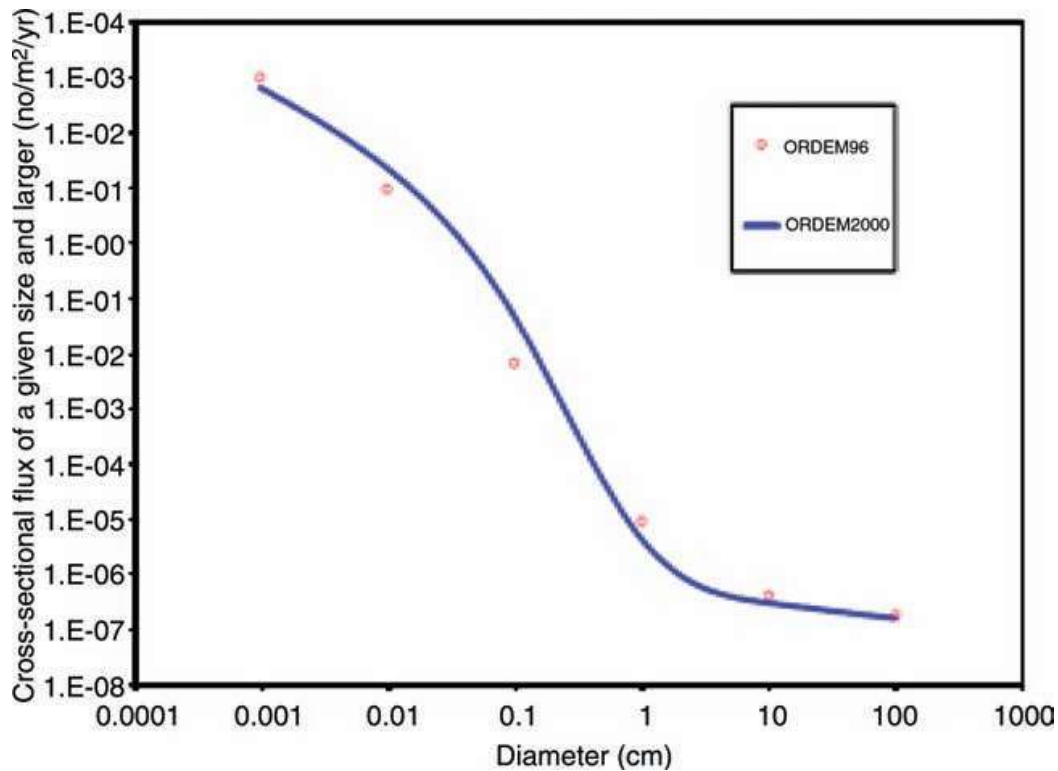


Fig. 11.7 ORDEM space debris flux for the International Space Station, inclination = 51.6 deg, eccentricity = zero, and altitude = 400 km in the year 2010. [From Liou et al. (2002); courtesy of NASA.]

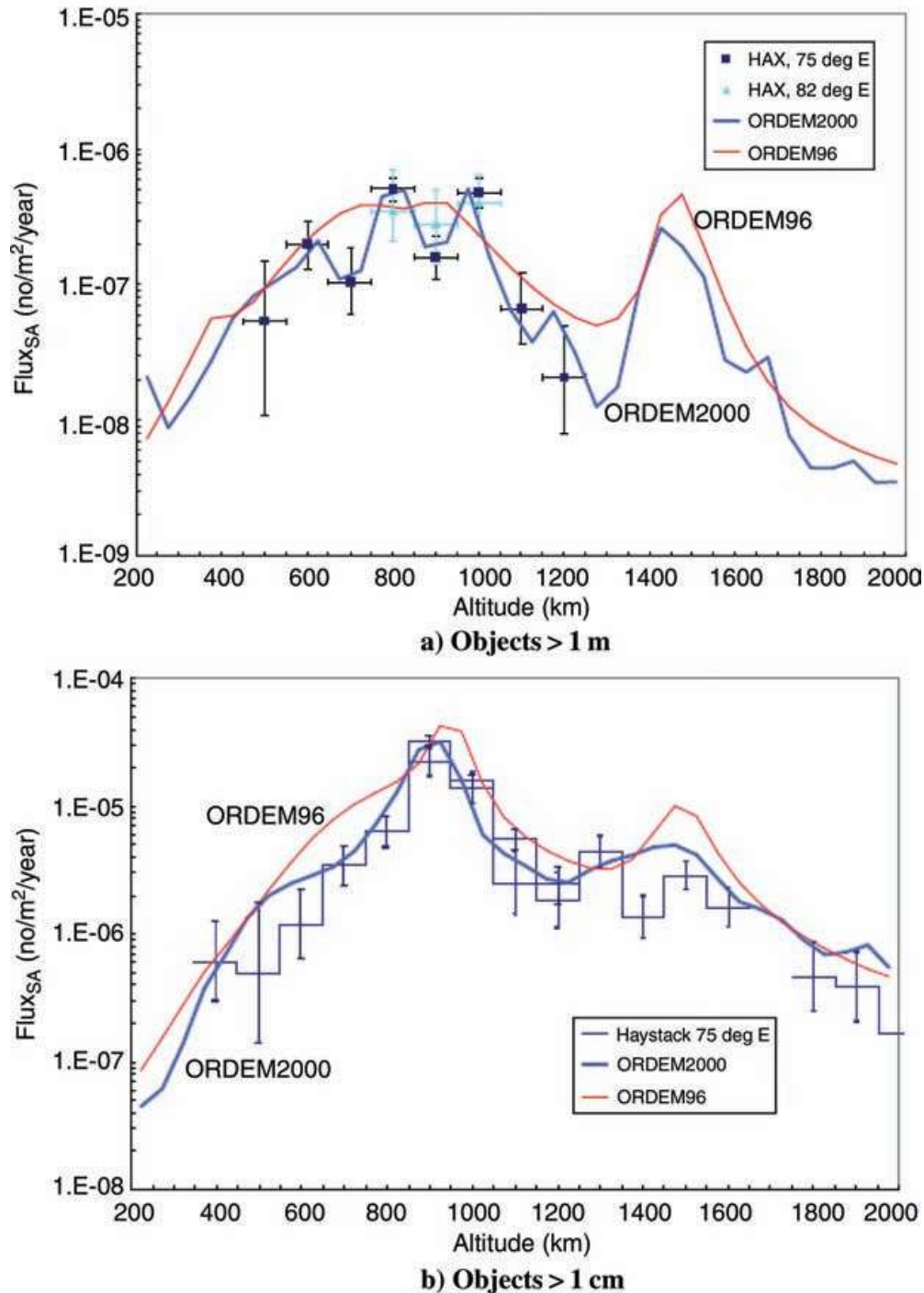


Fig. 11.8 Debris distribution as a function of altitude in ORDEM2000 and ORDEM96 for 1999. [From Liou et al. (2002); courtesy of NASA.]

of various sizes in low Earth orbit (LEO), medium Earth orbit (MEO), and geostationary Earth orbit (GEO) is given in Table 11.3.

Table 11.4 gives the time between impacts due to orbital debris as a function of altitude and particle size from the MASTER'99 model for a surface of area 100 m².

Table 11.3 Space debris population according to MASTER'99 model

Debris size	Objects in LEO	Objects in MEO	Objects in GEO
>0.1 mm	1.03×10^{10}	1.6×10^{11}	2.5×10^{10}
>1.0 mm	3.8×10^7	1.6×10^8	2.1×10^{10}
>1.0 cm	121,289	173,244	20,703
>10 cm	13,207	2,191	564

Source: From Walker et al. (2002); courtesy of the European Space Agency.

Figure 11.9 gives the density of debris particles from the MASTER 2001 model as a function of altitude and particle diameter. The density shows several prominent peaks. The peak at 900 km with period of 103 min orbit is thought to be due to the release of sodium-potassium liquid metal from the nuclear reactor of the Russian RORSAT ocean surveillance spacecraft. The peak at an altitude of 18,200 is due to rockets in geosynchronous transfer orbits with periods of 10.5 h. The nearby peak at an altitude of 20,200 km is due to spacecraft in semisynchronous orbits with periods of 11 h 58 min. The peak at an altitude of 35,780 km is due to spacecraft in geosynchronous orbits with periods of 23 h 56 min.

11.4.4 Debris Clouds

A debris cloud is a concentration of debris in a region of space and is formed when fragmentation occurs from a single object. If the fragmentation occurs from two objects, as in a collision, two debris clouds will form. Initially, a debris cloud will remain intact as it travels around the orbit, as illustrated in phase 1 in Fig. 11.10. However, due to differences in drag and or radiation pressure forces,

Table 11.4 Time between impacts for 100 m² surface area, from MASTER'99 model

Altitude, km		Time between impacts from debris objects per 100 m ² for particles greater than			
		0.1 mm	1 mm	1 cm	10 cm
400	Mean	4.5 days	3.9 yr	1,214 yr	16,392 yr
	Minimum	1.3 days	2.2 yr	776 yr	10,813 yr
	Maximum	33.1 days	5.9 yr	1,674 yr	21,853 yr
800	Mean	2.3 days	1.0 yr	245 yr	1,775 yr
	Minimum	1.3 days	0.6 yr	151 yr	1,198 yr
	Maximum	3.4 days	1.2 yr	310 yr	2,329 yr
1,500	Mean	0.9 days	1.5 yr	534 yr	3,109 yr
	Minimum	0.6 days	0.8 yr	323 yr	2,058 yr
	Maximum	1.4 days	2.0 yr	698 yr	4,305 yr

Source: From Walker et al. (2002); courtesy of the European Space Agency.

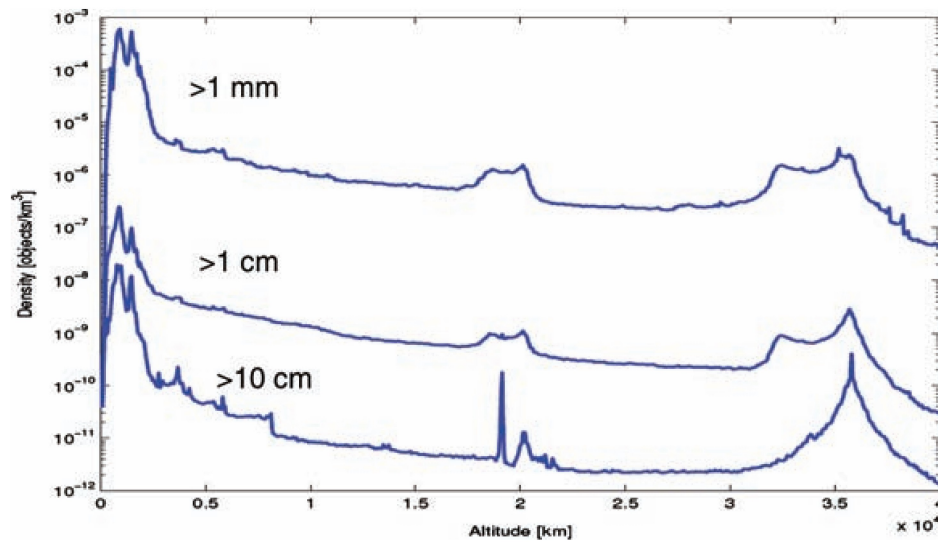


Fig. 11.9 Debris density from ESA MASTER 2001 model. [From Rossi (2005); courtesy of the International Astronomical Union.]

the debris cloud will spread out around the orbit into a toroid, as illustrated in phase 2 in Fig. 11.10. Due to slight changes in the inclinations of the debris, the right ascension of the ascending node of the orbit of each fragment will precess at a different rate and disperse, as illustrated in phase 3. Thus, debris clouds have a tendency to spread around the orbit and longitudinally.

11.4.5 Gabbard Diagram

Orbital clouds of space debris that result from a collision or a breakup are often assessed by use of a scatter plot known as a Gabbard diagram. The Gabbard diagram illustrates the dispersion of the apogee and perigee altitudes as a function of period for each fragment in the debris cloud. The distribution of apogee and perigee altitudes can be used to infer information such as direction and point of impact. Figure 11.11 is an example of the Gabbard diagram for the debris cloud of the fragmentation of the Abelstar upper stage on 29 June 1961. This is the first

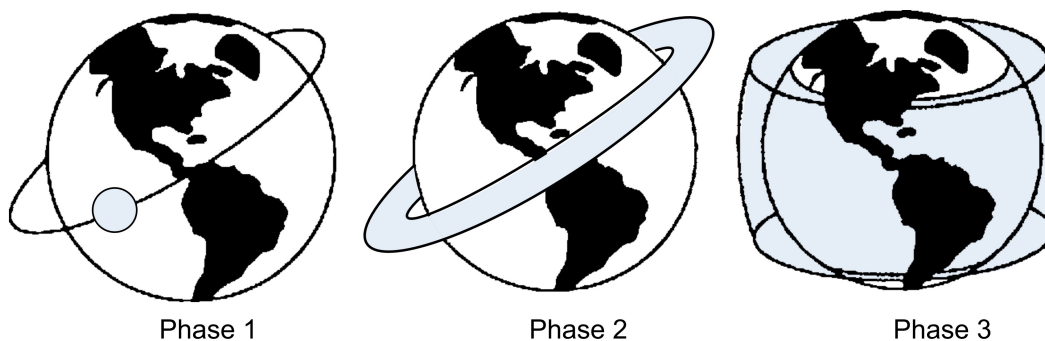


Fig. 11.10 Evolution of a debris cloud. (Courtesy Kaman Corporation.)

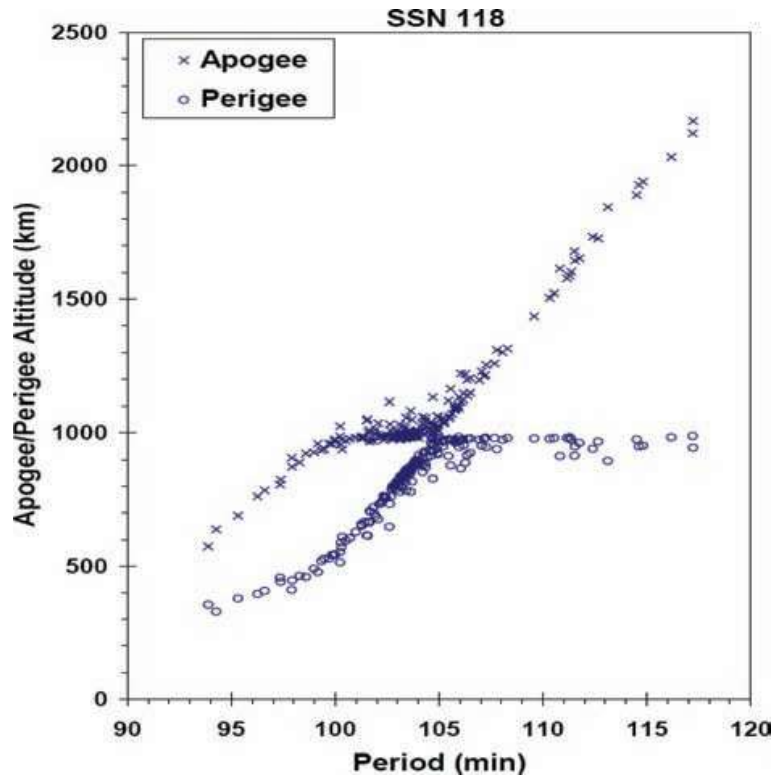


Fig. 11.11 Gabbard diagram for the Abelstar upper stage from U.S. SSN database of 201 cataloged fragments. [From Whitlock (2004); courtesy of NASA.]

known space fragmentation and resulted in 201 objects tracked by the U.S. Space Surveillance Network (SSN). At the time of the event, about 77 min after orbit injection of its three payloads, the Ablestar contained 100 kg of hypergolic propellants and the range safety explosive device.

11.5 Debris Mitigation

In the planning and execution of a space mission, it is important that measures are considered to minimize the adverse effects on the debris environment by an explicit debris mitigation plan. Considerations should address the following four areas: 1) debris produced during normal operations, 2) debris produced by explosions and intentional breakups, 3) debris produced by in-orbit collisions, and 4) post-mission disposal. Several Inter-Agency Space Debris Coordination Committee and NASA guidelines and standards are available to appropriately address minimizing debris: IADC-02-01 (IADC 2002), NASA NSS 1740.14 (NASA 1995), and NASA NPD 8710.3B (NASA 2004).

11.5.1 Release of Orbital Debris During Normal Operations

Planned release of orbital debris should be avoided. All lens caps, covers, clamps, etc., should be tethered. If release of debris cannot be avoided, then it is important that debris is minimized in number, area, and orbital lifetime.

11.5.2 *Debris Produced by Explosions and Intentional Breakups*

To minimize the potential of post-mission explosions, all sources of energy such as propellants, batteries, high-pressure vessels, self-destructive devices, flywheels and momentum wheels, etc., should be depleted or saved at the termination of the mission. Intentional destructions that produce debris that will remain in orbit for long periods of time should be avoided.

11.5.3 *Debris Generated by In-Orbit Collisions*

During the development of a space mission, the potential for in-orbit collisions should be minimized to reduce the potential for debris or damage that would not permit post-mission disposal.

11.5.4 *Post-Mission Disposal*

At the termination of the space mission, the spacecraft should be maneuvered to reduce the potential for collision.

For geosynchronous spacecraft, the recommendation is that the minimum perigee altitude be increased above the geosynchronous altitude of 35,800 km by

$$\Delta h_p = 235 + 1000 C_r (A/m) \quad (11.9)$$

where

(A/m) = area-to-mass ratio, $\text{m}^2 \text{kg}^{-1}$

C_r = radiation pressure coefficient, $1 \leq C_r \leq 2$

Δh_p = change in perigee altitude, km

The 235 km number is the sum of the upper altitude of the protected region of 200 km plus the luni-solar perturbation that lowers the perigee altitude by 35 km; see Fig. 11.12. A proposed alternative is to reduce the altitude by 500 km.

Another popular altitude is the semisynchronous orbit where the spacecraft has a period of half of a sidereal day. A proposed guideline is to increase or decrease the perigee altitude by 300 km, as illustrated in Fig. 11.12.

For spacecraft in low Earth orbit, several guidelines recommend that the orbit be changed to one that reenters within 25 years and would not pose a risk to people or property. Another proposal is to increase the altitude to at least 2500 km, where it will take many thousands of years for orbit decay to lower the altitude to <2000 km; see Figs. 11.1 and 11.12.

11.6 Collision Probabilities

Given the average debris or meteoroid flux, the Poisson distribution gives the probability of a collision in a specified time interval. In general, the Poisson distribution expresses the probability of the occurrence of a number of events that occur in a given time period if the events are independent and the average

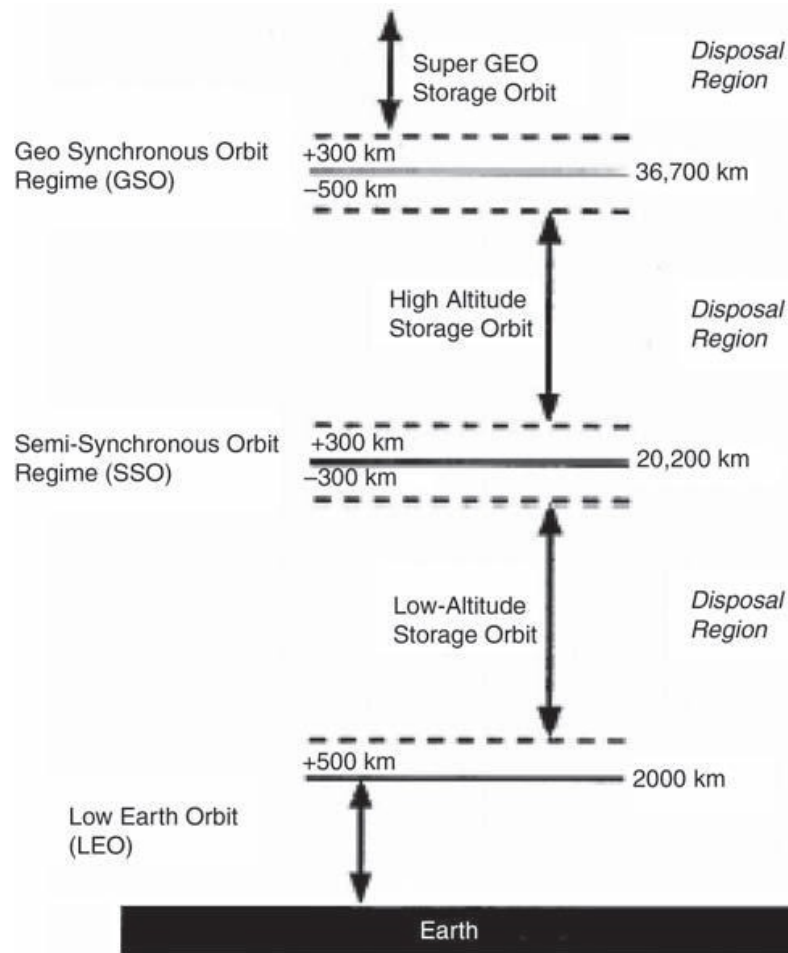


Fig. 11.12 Guidelines for post-mission disposal. [From NASA NSS 1740.14 (NASA 1995); courtesy of NASA.]

rate of the events is known. Mathematically, the Poisson distribution can be expressed as

$$p(k, \lambda t) = \frac{1}{k!} (\lambda t)^k e^{-\lambda t} \quad (11.10)$$

where

$p(k, \lambda t)$ = probability of k events given λt

k = number of events

λ = average number of events per unit time

t = time interval of interest

λt = shape parameter, average number of events in given time interval

The Poisson distribution is *memoryless*, which means that the number of events in any bounded interval of time after the time T is independent of the number of events occurring before the time T . The mean of the Poisson distribution is λt by

definition. It can be shown that the variance is also equal to λt , so that the standard deviation σ follows as

$$\sigma = \sqrt{\lambda t} \quad (11.11)$$

The probability of one or more events (collisions) is given by one minus the probability of no events (no collisions), so that

$$p(k \geq 1, \lambda t) = 1 - p(0, \lambda t) = 1 - e^{-\lambda t} \quad (11.12)$$

In terms of the flux density, it follows that

$$\lambda = F(m) A \quad (11.13)$$

where

A = area of interest

$F(m)$ = average flux density for mass $> m$, particles per unit time per unit area

so that Eq. (11.10) becomes

$$p(k, F(m) At) = \frac{1}{k!} (F(m) At)^k e^{-F(m) At} \quad (11.14)$$

For exposure to both meteoroids and space debris, the flux density is expressed as the sum, where

$$F_{m+d}(m) = F_m(m) + F_d(m) \quad (11.15)$$

so that Eq. (11.14) becomes

$$p(k, F_{m+d}(m) At) = \frac{1}{k!} (F_{m+d}(m) At)^k e^{-F_{m+d}(m) At} \quad (11.16)$$

The probability that there is one or more events (impacts) from debris and meteoroids follows from Eq. (11.12) as

$$\begin{aligned} p_{m+d}(k \geq 1, F_{m+d}(m) At) &= [1 - p_{m+d}(0, F_m(m) At)] \\ &= 1 - e^{-F_{m+d}(m) At} \end{aligned} \quad (11.17)$$

In the case that

$$F_{m+d}(m) At \ll 1 \quad (11.18)$$

then Eq. (11.17) can be approximated by

$$p_{m+d}(k \geq 1, F_{m+d}(m) At) \approx F_{m+d}(m) At \quad (11.19)$$

Example 11.1

Problem: A spacecraft with cross-sectional area of 1 m^2 is exposed to a micrometeoroid and debris environment where the average time to impact a particle with mass $>5 \times 10^{-4} \text{ g}$ is 0.1 years. Determine the probability of one or more impacts for 1 year, 5 years, and 10 years.

Solution: From the given information

$$F(m) = 0.1 \text{ m}^{-2} \text{ y}^{-1} \quad (11.20)$$

$$A = 1 \text{ m}^2 \quad (11.21)$$

Then Eq. (11.17)

$$p_{m+d}(k \geq 1, F_{m+d}(m)At) = 1 - e^{-F_{m+d}(m)At} \quad (11.17')$$

gives for 1, 5, and 10 years as follows:

For 1 year:

$$p_{m+d}(k \geq 1, F_{m+d}(m)At) = 1 - e^{-0.1 \times 1 \times 1} = 0.095 \quad (11.22)$$

For 5 years:

$$p_{m+d}(k \geq 1, F_{m+d}(m)At) = 1 - e^{-0.1 \times 1 \times 5} = 0.393 \quad (11.23)$$

For 10 years:

$$p_{m+d}(k \geq 1, F_{m+d}(m)At) = 1 - e^{-0.1 \times 1 \times 10} = 0.632 \quad (11.24)$$

11.7 Hypervelocity Impact Physics

Hypervelocity impacts are those that occur with relative speeds that exceed the speed of sound in the impacted material, which is typically around $4\text{--}5 \text{ km s}^{-1}$. Collisions with relative velocities greater than about $1\text{--}2 \text{ km s}^{-1}$ generally result in the impacting material behaving like a fluid. Relative velocity in low Earth orbit of two counter-rotating objects can be as high as 14 km s^{-1} and in geostationary orbit as high as 6 km s^{-1} . Impact velocities of meteoroids can be as high as 75 km s^{-1} . Figure 11.13 illustrates the diameters of aluminum spheres that produce the impact energy of a 30-caliber bullet and a hand grenade. Note that a 1-cm diameter sphere of aluminum with a relative velocity of 1 km s^{-1} has the impact energy of a 30-caliber bullet and at 55 km s^{-1} has an impact energy of the explosive force of a hand grenade. Figure 11.13 can be used for other than aluminum by scaling the energy by multiplying the energy by the density ratio ρ/ρ_{al} .

There are three primary means to assess and predict the potential damage to spacecraft from the impact of a hypervelocity particle: hypervelocity impact testing on Earth, analytical analyses and digital simulations, and in-situ observation

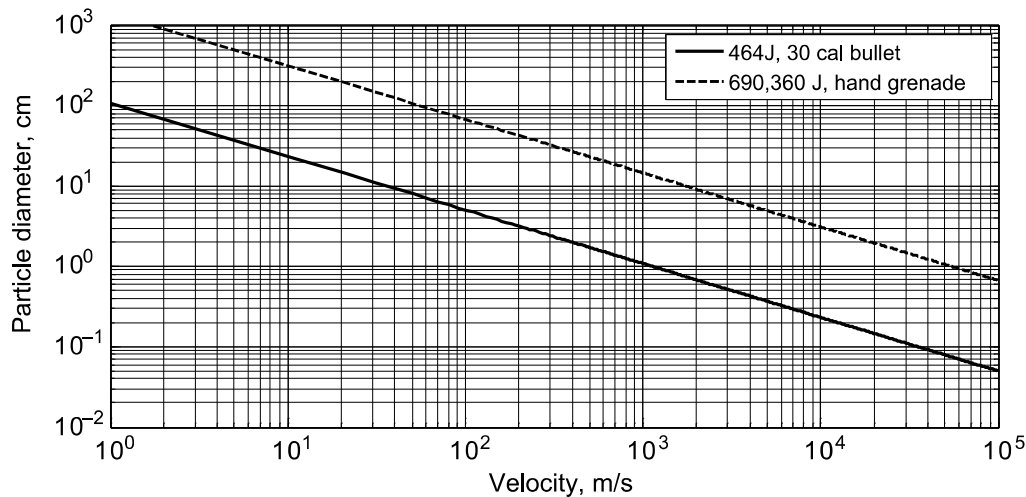


Fig. 11.13 Diameter of aluminum sphere for specified energy and velocity.

on returned space objects. Assessment of the potential damage of hypervelocity impacts is accomplished primarily through experimental testing and analytical or digital codes, as limited in-orbit data are primarily used for calibration and confirmation of the analyses.

For damage and risk assessment, the penetrability of a projectile into a structure is needed. A threshold equation for the penetration of a thin ductile plate by a projectile is given in NASA SP-8042 (NASA 1970), Fish and Summers (1965), and Sorensen (2006) as

$$t = k_1 m_p^{0.352} \rho_p^{0.167} v^{0.875} \quad (11.25)$$

where

- t = threshold thickness for penetration, cm
- k_1 = material constant, see Table 11.5
- m_p = mass of projectile, g
- ρ_p = density of projectile, g cm⁻³
- v = component of the relative velocity of the projectile normal to impacted surface, km s⁻¹

A penetration is deemed to occur when the threshold thickness for an impacting particle exceeds the shielding thickness of the surface impacted. Because most spacecraft surfaces are complex, such as honeycomb structures covered with multilayer insulation (MLI), the ability to predict penetration accurately is problematic. This is confounded further by the uncertainty in the mass, density, and relative speed of the anticipated projectile.

The maximum size of the diameter of the projectile that can be withstood by a thin plate, thick plate, MLI, and glass plate is given by (Space Environment Information System 2007)

$$d_{p,\text{lim}} = \left[t_t (K_r K_1 \rho_p^\beta \rho_t^\kappa v^\gamma \cos^\chi \alpha)^{-1} \right]^{1/\lambda} \quad (11.26)$$

Table 11.5 Values of k_1

Material	k_1
Aluminum alloys	
2024-T3	0.54–0.57
7075-T6	0.54–0.57
6061-T6	0.54–0.57
Stainless steel	
AISI 304	0.32
AISI 316	0.32
17-4 Ph annealed	0.38
Magnesium lithium	
LA 141-A	0.80
Columbium alloys	
Cb-1Zr	0.34

Source: From NASA SP-8042 (NASA 1970).

where

- $d_{p,\text{lim}}$ = projectile diameter limit, cm
- t_t = target thickness, cm
- K_f, K_1 = factors, dimensionless
- α = impact angle
- $\beta, \kappa, \gamma, \chi$ = parameters
- ρ_p = density of projectile, g cm^{-3}
- ρ_t = density of target, g cm^{-3}
- v = impact speed, km s^{-1}

Values of the parameters are given in Table 11.6.

Another model for penetration depth, crater diameter, and crater volume is given by Denardo et al. (1967) for samples of hard and soft aluminum.

11.8 Shields and Bumpers

The protection concepts for spacecraft from meteoroid and debris particles rely primarily on positioning metal shields or bumpers in front of critical surfaces in a variety of configurations. The Whipple shield (Whipple 1947) for meteoroids and debris consists of a thin shield in front of the wall structure to be protected. The purpose of the shield is to fracture the hypervelocity particle into a cloud of smaller particles consisting of the original projectile and shield material with some of the material possibly liquified and vaporized. The effectiveness of this approach depends on the ability of the shield to redistribute the momentum of the material cloud of particles over a larger area than it would otherwise impact without the shield. The Whipple shield generally has a mass advantage over single-walled structures. Whipple shields were used on the command module and lunar module of the Apollo missions. The Whipple shield is less effective at projectile velocities less than 3 km s^{-1} because the impact of a projectile with the shield may not cause it to fracture, liquefy, and vaporize sufficiently. Figure 11.14 illustrates the mechanics of an impact of a projectile with a Whipple shield.

Table 11.6 Parameters for Eq. (11.26)

Surface impacted	K_f^a	K_1^b	λ	β	γ	χ	κ
Thick plate	1.8–3.0	0.2–0.33	1.056	0.519	2/3	2/3	0
Thin plate	1.0	0.26–0.64	1.056	0.519	0.875	0.875	0
MLI	1.0	0.37	1.056	0.519	0.875	0.875	0
Glass	1.85–7.0	0.64	1.2	0	2/3	2/3	0.5
Glass	1.85–7.0	0.53	1.06	0.5	2/3	2/3	0

^a K_f

Thick plate	$3.0 \leq K_f$ $2.2 \leq K_f < 3.0$ $1.8 \leq K_f < 2.2$ $K_f < 1.8$	Crater without spall Spallation of the plate Spall breaks away Perforation of the plate
Thick glass	$7.0 \leq K_f$ $1.85 \leq K_f < 7.0$ $K_f < 1.85$	Crater without spall Spallation of the plate Perforation of the plate

^b K_1

Thick plate	Aluminum alloys Stainless steel	$K_1 = 0.33$ $K_1 = 0.20$
Thin plate	Aluminum alloys Stainless steel	$K_1 = 0.43\text{--}0.454$ $K_1 = 0.255$ AISI 304, AISI 306 $K_1 = 0.302$ AISI 17-4 PH annealed

Source: From Space Environment Information System (2007).

A number of derivatives of the Whipple shield have been studied that differ primarily by utilizing multiple shields with and without filling the intervening spaces with materials such as Kevlar®. Desirable characteristics of an effective shield are that it provides a low penetration risk to the primary structure with minimal mass and complexity of implementation. Shields can be optimized relative to the penetration of a specified object by selecting the shield materials, thickness, spacing, filling materials, and wall structure properties. Experimental and theoretical studies indicate that for typical objects and impact velocities in low Earth orbit that Whipple shields provide protection equivalent to monolithic shields 10–20 times their mass (Swift 1982).

Active protection systems can also be employed that utilize sensors to identify objects and their trajectories that could potentially impact a spacecraft or a launch vehicle. Onboard spacecraft sensors can be utilized with the detection range needed to be hundreds of kilometers so that objects at speeds greater than 10 km s^{-1} can be identified in time to take protective actions. The response could be the covering of sensitive components with protective shutters and maneuvering the spacecraft to avoid a collision. Potential launch trajectories are routinely evaluated for potential collisions with cataloged objects so that the launch can be scheduled to minimize risk. During human spaceflight, potential collisions with cataloged objects are routinely evaluated so that the astronauts can maneuver their spacecraft to avoid them. On the International Space Station, several hundred bumpers are employed that include louvers that protect the station's windows. Bumpers on the Apollo mission were capable of stopping a 0.16-cm-diam aluminum particle at a speed of 7 km s^{-1} at normal impact (Christiansen 2003).

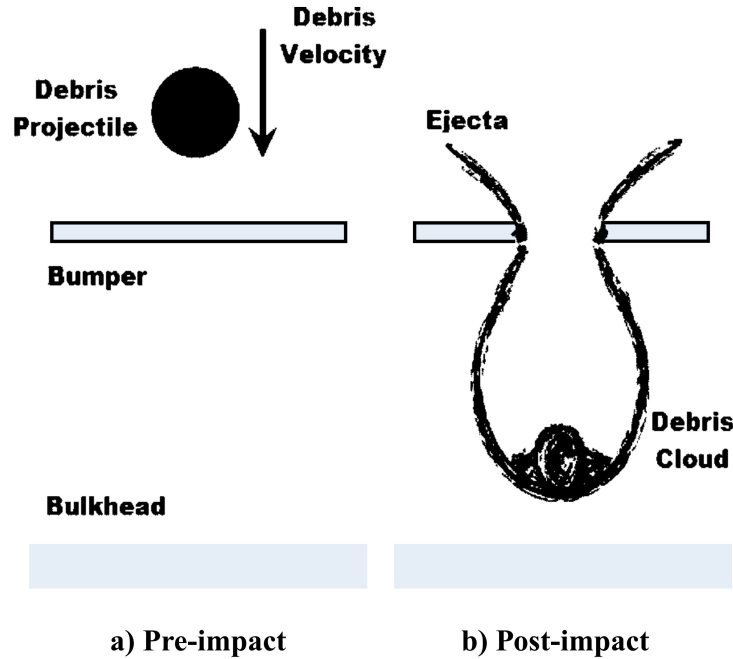


Fig. 11.14 Whipple shield mechanics.

Christiansen (2003) has developed design equations for the Whipple shield based on experimental data that can be used for conceptual designs when the velocity of the projectile is $\geq 7 \text{ km s}^{-1}$. The bumper thickness t_b and wall thicknesses t_w are given by

$$t_b = c_b d \frac{\rho_p}{\rho_b} \quad (11.27)$$

$$t_w = c_w d^{1/2} m_p^{1/3} (\rho_p \rho_b)^{1/6} \rho_w^{-1} S^{-3/4} \left(\frac{\sigma}{70} \right)^{-1/2} V \cos \theta \quad (11.28)$$

$$c_b = 0.25 \text{ for } S/d < 30 \quad \text{or } c_b = 0.20 \text{ for } S/d \geq 30 \quad (11.29)$$

$$c_w = 0.79 k \text{ cm}^{-3/4} \text{ g}^{1/3} \text{ km}^{-1} \text{ s} \quad (11.30)$$

$$k = \left(\frac{S/d}{15} \right)^{-0.185} \quad \text{for } S/d < 15 \text{ and } 1 \text{ otherwise} \quad (11.31)$$

where

- d = projectile diameter, cm
- m_p = projectile mass, g
- m_{pa} = projectile area density, g cm^{-2}
- S = spacing between shield and structure or standoff, cm
- t_b = bumper thickness, cm
- t_w = wall thickness, cm
- V = velocity of projectile, km s^{-1}
- ρ_b = bumper density, g cm^{-3}
- ρ_p = projectile density, g cm^{-3}

ρ_w = wall density, g cm^{-3}
 σ = wall yield stress, ksi (1000 psi)
 θ = impact angle measured from surface normal valid for $\leq 65^\circ$

Example 11.2

Problem: Determine the thickness required for the wall of a spacecraft if it has a Whipple shield to survive being struck by a spherical aluminum particle with a diameter of 1 cm at 10 km s^{-1} . Assume the shield and wall are made of aluminum alloy Al 6061-T6, whose yield stress is 35 ksi and density is 2.7 g cm^{-3} .

Solution: From the given information and for $V > 3 \text{ km s}^{-1}$ and $1 < S < 25$, it follows that

$$\begin{aligned} c_b &= 0.25 \\ c_w &= 0.79 \text{ km}^{-3/4} \text{ g}^{1/3} \text{ km}^{-1} \text{ s} \\ d &= 1 \text{ cm} \\ k &= \left(\frac{S/d}{15} \right)^{-0.185} \text{ for } S/d < 15 \text{ and } 1 \text{ otherwise} \end{aligned}$$

$$\begin{aligned} m_p &= (4\pi/3)(d/2)^3 \rho_p, \text{ g} \\ S &= \text{spacing between shield and structure, cm} \\ t_b &= \text{shield thickness, cm} \\ t_w &= \text{wall thickness, cm} \\ V &= 10 \text{ km s}^{-1} \\ \theta &= 0 \\ \rho_b &= \rho_p = \rho_w = 2.7 \text{ g cm}^{-3} \\ \sigma &= 35 \text{ ksi} \end{aligned}$$

The shield and wall thickness are given in Fig. 11.15 as a function of separation or standoff distance.

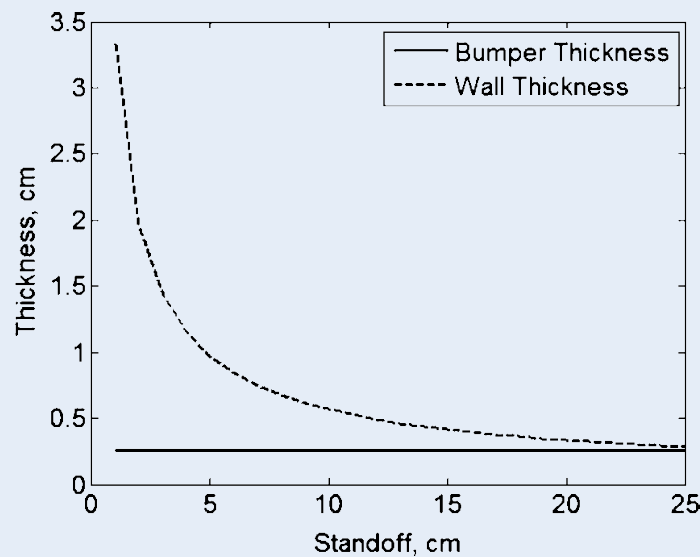


Fig. 11.15 Shield and wall thickness.

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Problems

- 11.1** Determine the speed of a 1-g micrometeoroid that is equivalent in energy to a 2-kg bowling ball dropped from 100 m.
- 11.2** Determine the following using the Grün et al. (1985) model for the meteoroid flux: (a) the flux for a particle with mass greater than 10^{-6} g, (b) ratio of the change in flux if the range of uncertainty in mass is a factor between 0.2 and 5.
- 11.3** Determine the following using the Grün et al. (1985) model for the meteoroid flux: (a) the flux for a particle with mass greater than 10^{-8} g, (b) ratio of the change in flux if the range of uncertainty in mass is a factor between 0.2 and 5.
- 11.4** Determine the fraction of meteoroids that have a velocity greater than 20 km s^{-1} .
- 11.5** Determine the fraction of meteoroids with a velocity less than 50 km s^{-1} .
- 11.6** Determine the allowed meteoroid and debris average flux for masses greater than a specified value for a 5-year mission that will result in a probability of 1 in a 10 of one or more impacts.
- 11.7** The average flux density for a spacecraft is $0.01 \text{ impacts m}^{-2} \text{ year}^{-1}$. For a spacecraft of area 1 m^2 and a period of 10 years, determine (a) the probability of no impacts, (b) the probability of one impact, and (c) the probability of two impacts.
- 11.8** Determine the threshold thickness for the penetration depth of a micrometeoroid with a mass of 1 g, density of 2 g cm^{-3} , and velocity of 10 km s^{-1} .